

- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Many bipolar patients may experience a reduction of symptoms by half or more
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Works –for Depression

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work –for Schizophrenia and Bipolar Disorder

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection

- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse

If It Doesn't Work –for Depression

- Some patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)



Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Other mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight

- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma

- Although US manufacturer recommends 6-month eye checks for cataracts, clinical experience suggests this may be unnecessary
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and quetiapine should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- By blocking histamine 1 receptors in the brain, it can cause sedation and possibly weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- Dose-dependent weight gain
- * May increase risk for diabetes and dyslipidemia
- * Dizziness, sedation
- Dry mouth, constipation
- Dyspepsia, abdominal pain
- Tachycardia
- Orthostatic hypotension, usually during initial dose titration
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- As a class, antidepressants have been reported to increase the risk of suicidal thoughts and behaviors in children and young adults

Weight Gain



- Many experience and/or can be significant in amount at effective antipsychotic doses
- Can become a health problem in some
- May be less than for some antipsychotics, more than for others

Sedation



- Frequent and can be significant in amount
- Some patients may not tolerate it
- More than for some other antipsychotics, but never say always as not a problem in everyone
- Can wear off over time
- Can reemerge as dose increases and then wear off again over time
- Not necessarily increased as dose is raised

What to Do About Side Effects

- Wait
- Wait
- Wait
- Usually dosed twice daily, so take more of the total daily dose at bedtime to help reduce daytime sedation
- Start dosing low and increase slowly as side effects wear off at each dosing increment
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 400–800 mg/day in 1 (quetiapine XR) or 2 (quetiapine) doses for schizophrenia
- 400–800 mg/day in 1 (quetiapine XR) or 2 (quetiapine) doses for bipolar mania
- 300 mg once daily for bipolar depression

Dosage Forms

- Tablet 25 mg, 50 mg, 100 mg, 150 mg, 200 mg, 300 mg, 400 mg
- Extended-release tablet 50 mg, 150 mg, 200 mg, 300 mg, 400 mg

How to Dose

- (According to manufacturer for quetiapine in schizophrenia): initial 25 mg/day twice a day; increase by 25–50 mg twice a day each day until desired efficacy is reached; maximum approved dose 800 mg/day
- In practice, can start adults with schizophrenia under age 65 with same doses as recommended for acute bipolar mania
- (According to manufacturer for quetiapine in acute bipolar mania): initiate in twice daily doses, totaling 100 mg/day on day 1, increasing to 400 mg/day on day 4 in increments of up to 100 mg/day; further dosage adjustments up to 800 mg/day by day 6 should be in increments of no greater than 200 mg/day
- Bipolar depression for quetiapine and quetiapine XR: once daily at bedtime; titrate as needed to reach 300 mg/day by day 4
- Quetiapine XR in schizophrenia and acute mania: initial 300 mg once daily, preferably in the evening; can increase by 300 mg/day each day until desired efficacy is reached; maximum approved dose 800 mg/day
- See also The Art of Switching section, after Pearls



Dosing Tips

- * **More may be much more:** clinical practice suggests quetiapine often underdosed, then switched prior to adequate trials
- Clinical practice suggests that at low doses it may be a sedative hypnotic, possibly due to potent H1 antihistamine actions, but this

can risk numerous antipsychotic-related side effects and there are many other options

* Initial target dose of 400–800 mg/day should be reached in most cases to optimize the chances of success in treating acute psychosis and acute mania, but many patients are not adequately dosed in clinical practice

- Many patients do well with immediate-release as a single daily oral dose, usually at bedtime
- Recommended titration to 400 mg/day by the fourth day can often be achieved when necessary to control acute symptoms
- Rapid dose escalation in manic or psychotic patients may lessen sedative side effects

* Higher doses generally achieve greater response for manic or psychotic symptoms

- In contrast, some patients with bipolar depression may respond well to doses less than 300 mg/day and as little as 25 mg/day
- Dosing in major depression may be even lower than in bipolar depression, and dosing may be even lower still in GAD

* Occasional patients may require more than 800–1200 mg/day for psychosis or mania

- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Quetiapine XR is controlled-release and therefore should not be chewed or crushed but rather should be swallowed whole
- Quetiapine XL may theoretically generate increased concentrations of active metabolite norquetiapine, with theoretically improved profile for affective and anxiety disorders
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Rarely lethal in monotherapy overdose; sedation, slurred speech, hypotension

Long-Term Use

- Approved for long-term maintenance in schizophrenia and bipolar disorder, and often used for long-term maintenance in various behavioral disorders
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- See also the Switching section of individual agents for how to stop quetiapine
- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Parent drug has 6–7 hour half-life
- Substrate for CYP450 3A4
- Food may slightly increase absorption



Drug Interactions

- CYP450 3A inhibitors and CYP450 2D6 inhibitors may reduce clearance of quetiapine and thus raise quetiapine plasma levels, but dosage reduction of quetiapine usually not necessary
- May enhance the effects of antihypertensive drugs
- May antagonize levodopa, dopamine agonists
- There are case reports of increased international normalized ratio (INR) (used to monitor the degree of anticoagulation) when

quetiapine is coadministered with warfarin, which is also a substrate of CYP450 3A4

Other Warnings/Precautions

- In the USA, manufacturer recommends examination for cataracts before and every 6 months after initiating quetiapine, but this does not seem to be necessary in clinical practice
- Quetiapine should be used cautiously in patients at risk for aspiration pneumonia, as dysphagia has been reported
- Priapism has been reported
- Use with caution in patients with known cardiovascular disease, cerebrovascular disease
- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Avoid use with drugs that increase the QT interval and in patients with risk factors for prolonged QT interval
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)

- Dysphagia has been associated with antipsychotic use, and quetiapine should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If there is a proven allergy to quetiapine

Special Populations

Renal Impairment

- No dose adjustment required

Hepatic Impairment

- Downward dose adjustment may be necessary

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Lower dose is generally used (e.g., 25–100 mg twice a day), although higher doses may be used if tolerated
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Approved for use in schizophrenia (ages 13 and older) and manic/mixed episodes (ages 10 and older)
- Clinical experience and early data suggest quetiapine may be safe and effective for behavioral disturbances in children and adolescents

- Children and adolescents using quetiapine may need to be monitored more often than adults and may tolerate lower doses better
- Use with caution, observing for activation of suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- When administered to pregnant rats and rabbits, embryofetal toxicity occurred at doses 1 and 2 times the maximum recommended human dose (MRHD), and maternal toxicity occurred at 2 times the MRHD in rats and 1–2 times the MRHD in rabbits

- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<https://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic>

Breast Feeding

- Unknown if quetiapine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on quetiapine should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- Bipolar depression
- Some cases of psychosis and bipolar disorder refractory to treatment with other antipsychotics
- * Patients with Parkinson's disease who need an antipsychotic or mood stabilizer
- * Patients with Lewy body dementia who need an antipsychotic or mood stabilizer

Potential Disadvantages

- Patients requiring rapid onset of action
- Patients who have difficulty tolerating sedation

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms
- Insomnia and anxiety

Pearls

- * May be the preferred antipsychotic for psychosis in Parkinson's disease and Lewy body dementia
- Anecdotal reports of efficacy in treatment-refractory cases and positive symptoms of psychoses other than schizophrenia
- * Efficacy may be underestimated for psychosis and mania since quetiapine is often underdosed in clinical practice
- * Approved in bipolar depression
- The active metabolite of quetiapine, norquetiapine, has the additional properties of norepinephrine reuptake inhibition and

antagonism of 5HT_{2C} receptors, which may contribute to therapeutic effects for mood and cognition

- Dosing differs depending on the indication, with high-dose mechanisms including robust blockade of D₂ receptors above 60% occupancy and equal or greater 5HT_{2A} blockade; medium-dose mechanisms including moderate amounts of NET inhibition combined with 5HT_{2C} antagonism and 5HT_{1A} partial agonism; and low-dose mechanisms including H₁ antagonism and 5HT_{1A} partial agonism and, to a lesser extent, NET inhibition and 5HT_{2C} antagonism
- More sedation than some other antipsychotics, which may be of benefit in acutely manic or psychotic patients but not for stabilized patients in long-term maintenance
- * Essentially no motor side effects or prolactin elevation
- May have less weight gain than some antipsychotics, more than others
- * Controversial as to whether quetiapine has more or less risk of diabetes and dyslipidemia than some other antipsychotics
- Commonly used at low doses to augment other atypical antipsychotics, but such antipsychotic polypharmacy has not been systematically studied and can be quite expensive
- Anecdotal reports of efficacy in PTSD, including symptoms of sleep disturbance and anxiety

- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma drug levels and, if low, a dosage increase even beyond the usual prescribing limits
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic

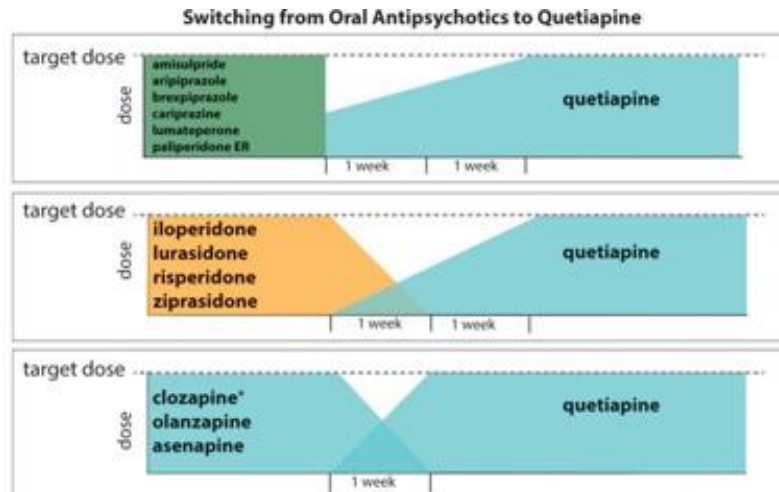
The Art of Switching

Switching from Oral Antipsychotics to Quetiapine

- With amisulpride, aripiprazole, brexpiprazole, cariprazine, lumateperone, and paliperidone ER, immediate stop is possible; begin quetiapine at middle dose
- With risperidone, ziprasidone, iloperidone, and lurasidone, it is generally advisable to begin quetiapine gradually, titrating over at least 2 weeks to allow patients to become tolerant to the sedating effect
- For more convenient dosing, patients who are currently being treated with divided doses of immediate-release tablets may be switched to extended-release quetiapine at the equivalent total daily dose taken once daily

* May need to taper clozapine slowly over 4 weeks or longer

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Suggested Reading

Citrome L. Adjunctive aripiprazole, olanzapine, or quetiapine for major depressive disorder: an analysis of number needed to treat, number needed to harm, and likelihood to be helped or harmed . *Postgrad Med* 2010 ;**122** (4):39 –48 .

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Suttajit S , Srisurapanont M , Maneeton N , Maneeton B . Quetiapine for acute bipolar depression: a systematic review and meta-analysis . *Drug Des Devel Ther* 2014 ;**8** :827–38 .

Ramelteon

Therapeutics

Brands

- Rozerem
- *see index for additional brand names*

Generic?

No

Class

- Neuroscience-based Nomenclature: melatonin receptor agonist (Mel-RA)
- Melatonin 1 and 2 receptor agonist

Commonly Prescribed for

(bold for FDA approved)

- **Insomnia (difficulty with sleep onset)**
- Primary insomnia
- Chronic insomnia
- Transient insomnia

- Insomnia associated with shift work, jet lag, or circadian rhythm disturbances

How the Drug Works

- Binds selectively to melatonin 1 and melatonin 2 receptors as a full agonist

How Long Until It Works

- Generally takes effect in less than an hour

If It Works

- Reduces time to sleep onset
- Increases total sleep time
- May improve quality of sleep

If It Doesn't Work

- If insomnia does not improve after 7–10 days, it may be a manifestation of a primary psychiatric or physical illness such as obstructive sleep apnea or restless leg syndrome, which requires independent evaluation
- Increase the dose
- Improve sleep hygiene
- Switch to another agent